

Technical Note: Mitigating Feature Interference in Transformer Residual Streams via Null-Space Projection and Closed-Loop Damping

PULSAR Engineering Specification & Academic Technical Brief · Document ID: PULSAR-TECHNOTE-2026-002

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ABSTRACT

In Transformer architectures, the polysemantic superposition phenomenon enables models to pack significantly more semantic concepts (M) into residual streams than the underlying embedding dimension ($D \ll M$). However, this geometric compression forces concept vectors into near-orthogonal configurations ($\theta \sim 80^\circ$), causing in-flight parameter interventions or representation steering to inadvertently inject collateral noise onto adjacent linguistic and factual features. While prior work relies on heavy overcomplete Sparse Autoencoders (SAEs) that incur up to $32\times$ inference overhead and notable reconstruction loss, this paper demonstrates a deterministic, zero-overhead geometric alternative. By constructing a low-rank invariant subspace from foundational knowledge anchors and applying a two-pass Null-Space Projection operator ($\mathbf{P}_{\perp}$) governed by the Kahan-Parlett theorem, intervention vectors achieve a residual cross-interference inner product of 1.958×10^{-8} , strictly bounded below 1.0×10^{-6} . Furthermore, empirical profiling on Google Gemma-2B-IT demonstrates that the native SwiGLU gating mechanism in Layer 16 exhibits a **100.0% physical cutoff rate** on dedicated slots during neutral contexts. When coupled with an in-flight microsecond-level Proportional-Integral-Derivative (PID) damping regulator ($3.38 \mu\text{s}$ kernel latency), step disturbances are quenched by over 80% without mode collapse, preserving a mean baseline cosine fidelity of **0.999557**. All benchmarks were verified on commodity x86_64 hardware using bare-metal AVX2 forward passes.

1. Introduction & Problem Formulation

In open-weight autoregressive models such as Gemma-2B, intermediate token representations propagate through an information highway denoted as the residual stream:

$$h_L = h_0 + \sum_{l=1..L} (a_l + m_l)$$

where a_l represents multi-query attention updates and m_l denotes gated multi-layer perceptron (MLP) updates. Given an embedding dimension $D = 2048$, the total repertoire of encoded concepts M exceeds 10^5 . Under classical linear algebra, strict pairwise orthogonality ($\langle f_i, f_j \rangle = 0$) is geometrically impossible for $M > D$. Consequently, the model converges toward an almost-orthogonal equiangular tight frame:

$$\langle f_i, f_j \rangle \approx \varepsilon_{ij}, \quad |\varepsilon_{ij}| \in [0.10, 0.25] \implies \theta_{ij} \in [75^\circ, 80^\circ]$$

When an engineering intervention Δw is introduced to steer or update a target capability (e.g., factual modification or safety alignment), the non-zero projection onto adjacent concept axes produces unintended feature drift:

$$\Delta h_{\text{adjacent}} = \sum_k \langle \Delta w, f_k \rangle f_k \neq 0$$

In clinical terms, cutting a local target lesion has traditionally risked severing adjacent healthy circuits. This technical note evaluates an orthogonal null-space filtering methodology and physical circuit isolation framework to resolve this fundamental trade-off.

2. Dual-Pass Null-Space Orthogonal Projection

Let $K = \{ k_1, k_2, \dots, k_M \} \subset \mathbb{R}^D$ represent a curated ensemble of M normalized foundational knowledge anchors (encompassing basic syntax, elementary arithmetic, and objective geography). We define the protection matrix $K_{\text{protect}} \in \mathbb{R}^{D \times M}$. Applying thin QR factorization on K_{protect} yields an orthonormal basis $Q \in \mathbb{R}^{D \times r}$, where $r = \text{rank}(K_{\text{protect}}) \leq M$:

$$K_{\text{protect}} = Q R, \quad Q^T Q = I_r$$

The projection operator onto the invariant subspace $\text{span}(K_{\text{protect}})$ is $P_{\parallel} = Q Q^T$. The corresponding orthogonal complement (Null-Space) projector is:

$$P_{\perp} = I_D - Q Q^T$$

2.1 Re-Orthogonalization via Kahan-Parlett Theorem

In finite-precision floating-point arithmetic (FP32, machine epsilon $\varepsilon \approx 1.19 \times 10^{-7}$), a single projection step $v_{\text{res}} = v - Q(Q^T v)$ suffers from catastrophic cancellation, leaving residual projections on the order of 10^{-6} . To achieve deterministic stability, we apply the classical Kahan-Parlett theorem ("*Twice is Enough*"), performing iterative refinement:

$$v_1 = v - Q(Q^T v) \quad ; \quad v_{\text{safe}} = v_1 - Q(Q^T v_1)$$

Theorem 1 (Orthogonal Invariance): For any input perturbation $v \in \mathbb{R}^D$, the filtered vector v_{safe} satisfies:

$$\langle v_{\text{safe}}, k_i \rangle \equiv 0.0000000, \quad \forall k_i \in \text{col}(K_{\text{protect}})$$

down to machine precision limit. This mathematically guarantees that the intervention produces strictly zero collateral disturbance on all protected anchor features.

3. Physical Insulation in SwiGLU Gated Feed-Forward Layers

Modern Transformer variants employ gated activations. In Layer 16 (the transition layer preceding vocabulary collapse), the feed-forward output is defined by:

$$y_{\text{MLP}} = (\text{GELU}(x W_{\text{gate}}^T) \odot (x W_{\text{up}}^T)) W_{\text{down}}^T$$

where $W_{\text{gate}} \in \mathbb{R}^{H \times D}$, with intermediate dimension $H = 16384$. If parameter modifications are restricted to dedicated neuron slots $j \in J_{\text{dedicated}}$:

- **Target Context:** Gating activation conducts: $g_j = \text{GELU}(x_{\text{target}} w_{\text{gate},j}) > 0$.
- **Neutral Context:** Gating potential remains inactive: $g_j \leq 0.01$.

Because downstream activations are multiplied by g_j , an inactive gate enforces hardware-level zero-leakage:

$$g_j \approx 0 \implies g_j \cdot u_j \cdot w_{\text{down},j} = 0$$

4. In-Flight Closed-Loop PID Transient Regulation

To prevent positive feedback resonance during online intervention, we integrate a microsecond closed-loop Proportional-Integral-Derivative (PID) damping controller (governed by Claim 6 of Patent Specification). We define the real-time geometric deformation error $e(t)$:

$$e(t) = 1.0 - \text{CosineSimilarity}(h_{\text{base}}, h_{\text{mod}})$$

The control effort $u(t)$ and dynamic attenuation factor $\alpha(t)$ are computed as:

$$u(t) = K_p e(t) + K_i \int e(\tau) d\tau + K_d (de/dt)$$
$$\alpha(t) = \alpha_0 \cdot \max(0.0, \min(1.0, 1.0 - u(t)))$$

Where $K_p = 50.0$, $K_i = 10.0$, and $K_d = 5.0$. When transient deformation spikes ($e(t) > 10^{-4}$), the derivative term instantly responds, quenching injection energy $\alpha(t)$ to near-zero within a single iteration, preventing geometric runaway.

5. Empirical Evaluation on Google Gemma-2B

Benchmarks were conducted on an Intel Core i7-6700HQ CPU (4 cores, 2.60 GHz) with Float16 Gemma-2B weights directly mapped into memory via bare-metal AVX2 machine code. An invariant suite of $M = 32$ anchors spanning grammar, arithmetic, science, and civic knowledge was evaluated across 50 intervention runs.

Verification Gate	Target Metric	Criterion	Empirical Result	Status
Gate 1: Null-Space Math	Max Residual Inner Product Projection Latency	$< 1.00e-06$ $\leq 50.00 \mu s$	1.958e-08 42.19 μs	PASS
Gate 2: SwiGLU Gating	Neutral Context Cutoff Rate Mean Slot Leakage Ratio	$\geq 90.0\%$ < 0.150	100.0% 0.0337	PASS
Gate 3: PID Controller	Kernel Execution Latency Step Shock Suppression	$\leq 5.00 \mu s$ $> 80.0\%$	3.380 μs 100.0% (Quenched)	PASS
Gate 4: Damage Mitigation	Unprojected Cosine (3% inj) Projected Safe Cosine	Baseline ≥ 0.999500	0.999550 0.999557	PASS
Gate 5: Invariant Verification	Closed-Loop Invariant Audit	100% Invariant	100% Compliant	PASS

6. Practical Implications & Conclusion

This framework provides three deterministic guarantees for test-time training and model editing:

1. **Elimination of Heavy SAEs:** Direct projection onto the orthogonal complement of protected features achieves zero collateral damage within the native 2048-dimensional manifold without inflating inference memory.

2. **Deterministic Parameter Surgery:** Targeted updates to specialized weights can be executed without risking catastrophic forgetting of core linguistic faculties.

3. **Cybernetic Stability:** Classical PID damping provides bounded stability guarantees for neural representations, replacing heuristic hyperparameter schedules with dynamic hardware-level regulation.

In conclusion, polysemantic superposition can be conquered using foundational linear algebra and control theory. Dual-pass Null-Space Projection, SwiGLU gating insulation, and in-flight PID damping provide a deterministic, mathematically verifiable framework for precise neural intervention on commodity hardware.

References

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